IKRAMUL HASAN MORAL

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EDUCATION

B.Sc. in Computer Science and Engineering | United International University (UIU), Dhaka — Expected Jan/Feb 2027
CGPA: 3.94/4.0 | 8 merit scholarships: 6 at 100% (top 3% each trimester), 1 at 50%, 1 at 25%.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, C++, Java, SQL, HTML/CSS
Frontend: React, Next.js, React Native, Expo, Tailwind CSS, Radix UI, Zustand, TanStack Query, Redux
Backend: Node.js, Express, NestJS, MongoDB, PostgreSQL, Redis, Supabase, Firebase, REST/OpenAPI, JWT, Socket.io, WebRTC
Machine Learning: PyTorch, scikit-learn, XGBoost, SMOTE/imbalanced-learn, pandas, NumPy, SciPy, Matplotlib
Tools: Git, GitHub Actions, Docker, Vercel, Railway, Linux, Postman, Stripe, Cloudinary, Arduino

EXPERIENCE

Undergraduate Teaching Assistant | United International University, CSE — Aug 2025 - Present
- Assist with 4 labs (DBMS, DSA-I, ICS, SWE), covering SQL, normalization, graph algorithms, and dynamic programming.
- Write and grade programming assignments. Hold weekly office hours and review student code.

Software Engineer | RimonsIELTS | Sole Engineer, Remote — 2025 - Present
- Maintain 4 live apps: Next.js 15 portal, Express/TypeScript REST API, React 19 admin dashboard, and Python service.
- Rebuilt the 2022 MERN site. Added Vimeo/Zoom, AI study tools, file storage, and course and exam management.
- Maintain Zod input checks, OpenAPI docs, JWT authentication, server tests, and release checklists.

RESEARCH

Lorenz-1960 Physics-Informed Neural Network | Undergraduate Thesis | Code — 2025 - 2026
- Led a team of 5 supervised by Dr. Muhammad Nomani Kabir. FYDP-II defended; manuscript submitted to ICEEICT 2027.
- Trained a residual-only PINN (11,283 parameters) in under 6 CPU minutes, within 3e-05 of the DOP853 reference. Tested 9 architectures and archived 32 models with SHA-256 verification and 41 tests.
- **UNSW-NB15 Class Imbalance (2025 - 2026, term paper):** compared LR, RF, and XGBoost with class weights and SMOTE using fixed seeds and leakage checks. Measured macro-F1, ROC-AUC, and G-Mean.
- **DT-NB (2026):** compared Naive Bayes instance filtering and tree-based attribute weighting within training folds across 10 seeds (Weka J48). Manuscript written, not yet submitted.
- **CAN Attack Detection (2026):** study physics-residual vs. data-driven attack attribution on the ROAD dataset.
- **Hybrid Nanofluid Flow (2026):** trained 9-layer ANN on ~32,400 physics-generated samples using Levenberg-Marquardt; validated against numerical solutions (MSE, RMSE, R-squared).

PROJECTS

Advyon - Legal-Tech Platform | React, Express, TypeScript, MongoDB | GitHub | Live — 2025
- Led a team of 5 from requirements to delivery. Built case and document management, intake forms, a forum, and AI-assisted legal insights with JWT authentication and role-based access.

BriefLi - AI Shortlink Manager | React, TypeScript, Supabase, Gemini API | GitHub | Live — 2025 - Present
- Added AI slug suggestions, click tracking, public and account-based links, and trash recovery. Built a Vercel API with Supabase PostgreSQL migrations and unique-slug constraints.

Fall Detection Wearable | Arduino/C++, Python ML, Node.js, JavaScript | GitHub — 2025
- Connected wearable firmware to a Python fall classifier, Node.js alert server, and caregiver dashboard to detect falls from live accelerometer readings.

Additional projects: Shera Bot, VoiceKey, Paradox Prism, PrintSlot, Carried, ODE 3D Orbit, Skill Paradox, Meme Research Console, Email Templates.

ACHIEVEMENTS & LEADERSHIP

- **Champion:** UIU CSE Project Show, Software Engineering Lab (Advyon, team lead); CSE Week Programming Contest 2024 (80+ participants).
- Solved 500+ problems on Codeforces, LeetCode, and HackerRank (2022 - 2025).
- 1st Runner-Up, Advanced OOP Project Showcase 2023; 6th Runner-Up, System Analysis & Design Lab Showcase 2024.
- Peer mentor, UIU CSE community; created exam study packs used across the department.